

Supplemental Material

“Deep Learning–Based Continuous QT Monitoring to Identify High-Risk Prolongation Events After Class III Antiarrhythmic Initiation”

Detailed Methods

Inter-observer Variability Analysis

Clinical ECGs were each interpreted by a single reader. Therefore, a direct inter-observer agreement statistic (such as Cohen’s Kappa) could not be calculated. To estimate the contribution of reader bias toward total QT measurement variability we performed a mixed-effects model variance decomposition as follows:

- Software and packages: Analyses were conducted in Python 3.11 using statsmodels (v0.14.0), pandas (v2.2.2), and numpy (v1.26.4).
- A random intercept mixed linear model was fit across QT intervals with reader as the random effect.
- Models were fit using restricted maximum likelihood (REML = True)
- Between-observer variance represented the inter-observer bias while within-observer variance reflects the physiological and clinical variability.
- Variance ratio was computed as the ratio of the between- and inter-observer variances. Proportion of variance is equal to the ratio of the between- and the sum of the between- and inter-observer variances.

Development of the 3DRECON-QT Model

The 3DRECON-QT model utilizes an encoder-decoder framework trained via a multi-task learning objective. This design forces the model to learn a physiologically grounded latent representation by simultaneously reconstructing the full 12-lead ECG and predicting the scalar QT interval from a single input lead.

The overall architecture takes in an input of a 10-second single-lead ECG waveform, sampled at 500Hz. The model employs an encoder (SE-ResNeXT) that distills the input signal into a latent representation. Two separate heads operate simultaneously, one for 12-lead ECG reconstruction, and one for QT interval prediction (**Figure 2**). The reconstruction head synthesizes the standard 12-lead ECG by treating each lead as a unique spatial ‘viewpoint’ of the heart’s electrical activity with respect to the thorax. A ThetaEncoder first generates a spatial embedding from a target ECG lead’s position, mathematically represented as a spherical coordinate pair (θ, ϕ) . The spatial information is then fused with the latent representation of the input single-lead ECG, effectively conditioning it with the information of the target viewpoint. An ECGDecoder then translates this conditioned representation into the corresponding waveform. The QT prediction head bypasses spatial conditioning and instead uses the complete latent representation to preserve available spatiotemporal information. This latent representation is then fed into a TransformerDecoder, which uses self-attention to identify salient features and their relationships across the cardiac cycle for a globally-informed QT assessment. The

resulting embedding is passed to a final Multi-Layer Perceptron (MLP) regressor to output the scalar QT prediction.

The model uses a weighted combination of two loss functions, one for the reconstruction and one for the QT regression. Optimization details included a stochastic gradient descent optimizer with an initial learning rate of $1e-3$ with cosine annealing with warm restarts. The regularization was L2 weight decay of $1e-5$ and batch size of 128. To prevent overfitting, the model was trained for up to 100 epochs with an early stopping protocol. The final checkpoint was selected based on the best Pearson correlation for QTc prediction on the validation set. The model was implemented in Python using the PyTorch framework (v2.3.1).

Supplemental Tables

Table S1: Data Availability and Model Input by Cohort

Cohort	Data Available	Model Input	12-Lead ECG Use
Stanford Model Development	12-lead ECG	V3-V2 derived signal	Training loss & QT validation
Cedars-Sinai External Validation	12-lead ECG	V3-V2 derived signal	QT validation
PhysioNet ECGRDVQ	12-lead ECG	V3-V2 derived signal	QT validation
Stanford Class III Antiarrhythmic Exposure	12-lead ECG	V3-V2 derived signal	QT validation
Stanford ICM Recording	Real ICM + 12-lead ECG	Real ICM signal	QT validation

Table S2: ECG-wise patient characteristics

	Train (n=637,589)	Validation (n=78,979)	Test (n=78,245)	External (n=100,000)
Age	61.8 ± 17.5	62.2 ± 17.3	62.0 ± 17.4	64.5 ± 17.7
Male	52.3%	52.4%	51.6%	55.0%
Race				
Asian	13.3%	13.6%	13.8%	7.2%
Black	5.7%	6.0%	5.7%	15.7%
Native American	0.3%	0.3%	0.3%	0.3%
Pacific Islander	1.7%	1.4%	1.3%	0.4%
White	57.6%	57.3%	57.6%	66.9%
Other	16.5%	16.3%	16.5%	7.0%
Unknown	1.2%	1.3%	1.1%	2.6%
Ethnicity				
Hispanic/Latino	12.7%	12.2%	13.4%	14.5%
Non-Hispanic	82.2%	82.5%	81.5%	82.7%
Unknown	1.2%	1.3%	1.1%	2.8%
AT/AF	28.7%	28.4%	28.2%	3.6%
CAD	24.7%	25.3%	24.3%	28.6%
HTN	53.0%	53.6%	53.1%	50.3%
HF	23.7%	23.0%	23.0%	26.3%

AT/AF - atrial tachycardia or atrial fibrillation, CAD – coronary artery disease, HTN – hypertension, HF – heart failure

Table S3: Performance metrics with bias-corrected confidence intervals

Metric	Stanford	Stanford 95% CI	Skewness	Cedars	Cedars 95% CI	Skewness
<i>Regression</i>						
MAE	17.522	17.346–17.699	-0.171	21.137	21.006–21.274	0.094
R ²	0.747	0.741–0.753	0.047	0.738	0.733–0.743	0.057
Pearson r	0.864	0.861–0.868	0.041	0.859	0.856–0.862	0.051
<i>Classification</i>						
Sensitivity	0.599	0.587–0.614	0.270	0.542	0.533–0.549	-0.170
Specificity	0.977	0.976–0.978	0.050	0.988	0.988–0.989	-0.009
Precision	0.605	0.593–0.618	0.150	0.883	0.877–0.889	-0.033
NPV	0.977	0.976–0.978	0.193	0.929	0.927–0.930	0.071
Accuracy	0.956	0.955–0.958	0.058	0.925	0.923–0.926	0.109
F1 Score	0.602	0.592–0.614	0.237	0.672	0.664–0.678	-0.184
<i>AUC</i>						
ROC AUC	0.941	0.938–0.945	0.094	0.943	0.941–0.945	0.038
AUPRC	0.633	0.621–0.648	0.206	0.781	0.775–0.787	-0.034

MAE – mean absolute error, NPV – negative predictive value, AUROC – area under the receiver-operating characteristic curve, AUPRC – area under the precision-recall curve

Table S4: 12-lead reconstruction performance

To assess reconstruction performance, we used the Pearson correlation coefficient. For each individual lead in the standard 12L ECG, we compared the predicted waveform from 3DRECON-QT against the original ground truth. The Pearson correlation was calculated between these two signals to get a score measuring their morphological similarity. We ran this analysis across the testing split of the Stanford Model Development Cohort, collecting and reporting both the lead-wise metric and mean score across leads.

Metric	I	II	III	aVR	aVL	aVF	V1	V2	V3	V4	V5	V6	Mean
Pearson r	0.71	0.76	0.41	0.78	0.53	0.60	0.74	0.74	0.76	0.78	0.79	0.78	0.70

Table S5: Patient characteristics for the Stanford Real-world Class III Antiarrhythmic Exposure Cohort

	Class III AAD Cohort (N=1,676)
Age	68.2 ± 14.1
Creatinine (mg/dL)	1.1 ± 0.5
Potassium (mEq/L)	4.1 ± 0.5
Magnesium (mg/dL)	2.0 ± 0.3
AT/AF	1272 (75.9%)
HF	496 (29.6%)
Drug Class	
Dofetilide	332 (19.8%)
Sotalol	1344 (80.2%)

AT/AF - atrial tachycardia or atrial fibrillation, HF – heart failure

Supplemental Figures

Figure S1

The distribution for QT intervals per reader across the top 20 readers (anonymized Reader IDs) by ECG count in the dataset. The variability in the mean and QT label distributions between readers is small compared to the overall variability in the QT measurements.

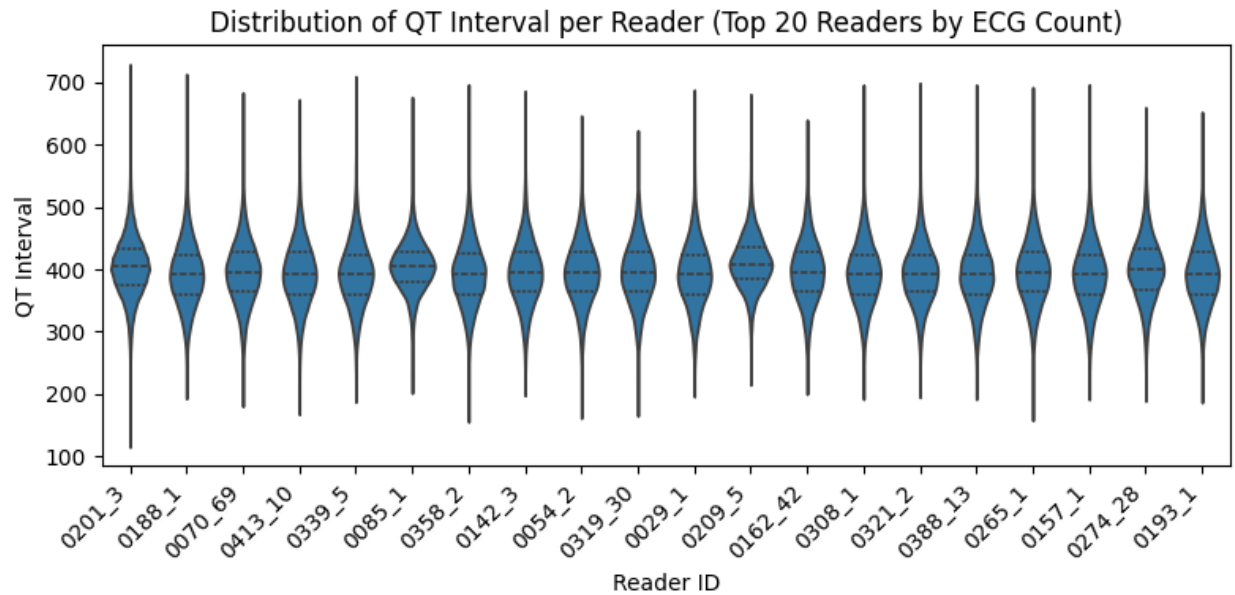


Figure S2

To generate lead-wise saliency maps, we first derived a global feature importance vector from the attention weightings within the model's QT decoder. This vector quantifies the importance of different features in the shared latent space for the overall QT prediction. We then projected the importance-scaled latent features onto the learned spatial embeddings for each lead (blue, ground truth 12-lead ECG, dashed red, 12-lead reconstruction). This process created a distinct temporal saliency map for every lead, highlighting the specific time points the model found most influential. Finally, these saliency maps were Gaussian smoothed to identify top regions of interest, which were visualized with bounding boxes (yellow). The boundary for each box was defined by the area around each peak where the saliency score remained highest in attention weightings. Notably, the activations for QRS onset and T wave offset vary by lead, emphasizing the spatial basis of the QT predictions requiring earliest onset of QRS and latest offset of T wave.

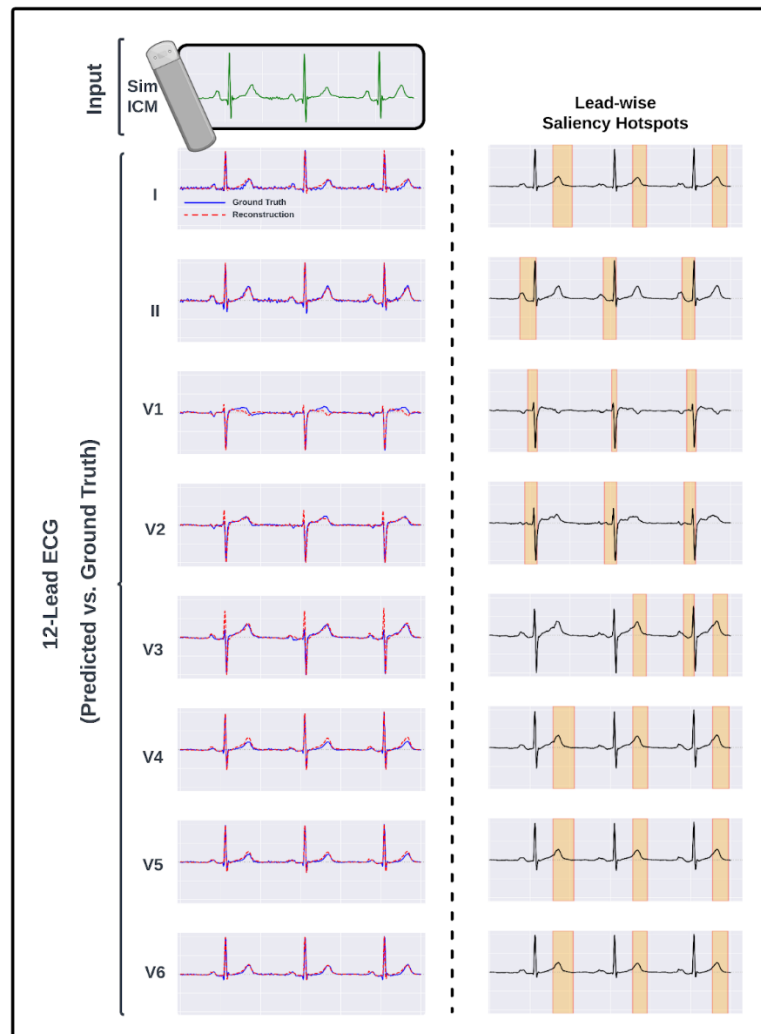


Figure S3

To evaluate the contribution of the spatial encoding to QT prediction, we compared 3DRECON-QT to an ablated variant. Both models were trained on identical data from the Stanford Model Development Cohort and shared an identical encoder, feature extraction, and regression architectures, with weights initialized on the same random seed. The variants differed only in the presence/absence of the core spatial encoding functionality, i.e., the theta encoder, reconstruction head, and reconstruction loss objective. To characterize generalization performance over time, validation metrics (Pearson correlation, MAE) were computed on held-out data at initialization and every 100 training steps. The training loss was recorded and reported with respect to the weighted MSE loss for the QT regression task. The model with spatial encoding achieved QTc MAE of 25.9 ms and Pearson r of 0.73, while the ablated model without spatial encodings showed markedly degraded performance (QTc MAE: 39.23 ms, Pearson r : 0.04), despite similar training loss convergence. This demonstrates the spatial pathway provides essential physiological constraints rather than merely aiding optimization.

